

2.3

13 $\lim_{x \rightarrow 2} \frac{x-2}{x^3-8}$

$\left. \begin{array}{c} \text{ } \\ \text{ } \end{array} \right\} \lim_{x \rightarrow 2} \left(\frac{\overbrace{x-2}^{f(x)}}{x^3-8} \right)$

$x \rightarrow 2 \Rightarrow x \approx 2 \Rightarrow f(x) \approx f(2) = \frac{2-2}{2^3-8} = \frac{0}{0} \quad (!?)$
 INDET.

DIVIDENDO

$\overline{) x^3 - 8} \quad | \quad \overline{) x - 2} \leftarrow \text{DIVISOR}$

$\underline{-x^3 + 2x^2} \quad \underline{x^2 + 2x + 4} \leftarrow \text{QUOCIENTE}$

$\underline{2x^2 - 8}$
 $\underline{-2x^2 + 4x}$
 $\underline{4x - 8}$
 $\underline{-4x + 8}$

$0 \leftarrow \text{Resto}$

DIVIDENDO = QUOCIENTE $\times$ DIVISOR + RESTO

$x^3 - 8 = (x^2 + 2x + 4) \cdot (x - 2) + 0$

$x^3 - 8 = (x^2 + 2x + 4) \cdot (x - 2)$

Logo, $\underline{f(x)} = \frac{x-2}{x^3-8} = \frac{\cancel{x-2}}{(x^2+2x+4) \cdot \cancel{(x-2)}}$

$= \frac{1}{x^2+2x+4}$

$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \left(\frac{1}{x^2+2x+4} \right) = \frac{1}{2^2+2 \cdot 2+4}$
 $= \frac{1}{4+4+4} = \frac{1}{12}$

14 $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{(x-2)^2}$

$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{(x-2)^2}$

$$\begin{array}{r}
 x^2 - x - 2 \\
 \underline{-x^2 + 2x} \\
 x - 2 \\
 \underline{-x + 2} \\
 0
 \end{array}
 \quad \bigg| \begin{array}{l} (x-2) \\ x+1 \end{array}
 \Rightarrow x^2 - x - 2 = (x-2)(x+1)$$

Logo,

$$f(x) = \frac{x^2 - x - 2}{(x-2)^2} = \frac{\cancel{(x-2)} \cdot (x+1)}{\cancel{(x-2)} \cdot (x-2)}$$

$$f(x) = \frac{x+1}{x-2}$$

$$\lim_{x \rightarrow 2} \frac{x+1}{x-2}$$

$$\begin{array}{lcl}
 x \rightarrow 2 \Rightarrow x \approx 2 & \rightarrow & 2,000\dots1 \xrightarrow{x \rightarrow 2^+} \frac{2,000\dots1 + 1}{2,000\dots1 - 2} \approx \frac{3}{0,00\dots1} = +\infty \\
 & \rightarrow & 1,999\dots9 \xrightarrow{x \rightarrow 2^-} \frac{1,999\dots9 + 1}{1,999\dots9 - 2} \approx \frac{3}{-0,000\dots1} = -\infty
 \end{array}$$

Ou seja,

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \text{e} \quad \lim_{x \rightarrow 2^-} f(x) = -\infty$$

Limites laterais existem, mas são $\neq$
 Portanto NÃO existe $\lim_{x \rightarrow 2} f(x)$

$$T. \lim_{x \rightarrow c} (ax+b) = a \cdot c + b$$

$$18 \quad \lim_{x \rightarrow \pi} (x - 3,1416)$$

$$\begin{array}{l}
 \Rightarrow \lim_{x \rightarrow \pi} (x - 3,1416) = \pi - 3,1416 < 0 \\
 \begin{array}{l} a = 1, b = -3,1416 \\ c = \pi \end{array} \\
 \pi = 3,1415926\dots
 \end{array}$$

25 $\lim_{h \rightarrow 0} \frac{4 - \sqrt{16+h}}{h}$

"TÉCNICA

DO CONJUGADO"

$$\frac{4 - \sqrt{16+h}}{h} \cdot \frac{(4 + \sqrt{16+h})}{(4 + \sqrt{16+h})}$$

Obs: $\sqrt{a^2} \neq a$
 $\parallel$
 $|a|$

$$= \frac{16 + 4\sqrt{16+h} - 4\sqrt{16+h} - \underbrace{(16+h)}_{h \rightarrow 0 > 0 = 16+h}}{h(4 + \sqrt{16+h})}$$

$$= \frac{16 - (16+h)}{h(4 + \sqrt{16+h})} = \frac{-h}{h(4 + \sqrt{16+h})}$$

$$= \frac{-1}{(4 + \sqrt{16+h})} \quad f(h)$$

Logo,

$$\lim_{h \rightarrow 0} f(h) = \frac{-1}{4 + \sqrt{16+0}} = -\frac{1}{8}$$

32 $\lim_{t \rightarrow 7} \frac{\sqrt[5]{3-5t}}{(t-5)^3} \quad f(t)$

$$t \rightarrow 7 \Rightarrow t \approx 7 \Rightarrow f(t) \approx \frac{\sqrt[5]{3-5 \cdot 7}}{(7-5)^3} = \frac{\sqrt[5]{-32}}{2^3} = \frac{-2}{8} = -\frac{1}{4}$$

36 $\lim_{h \rightarrow 0} \frac{(9+h)^2 - 9^2}{h} = \frac{1}{\frac{9+h}{9+h} - \frac{1}{9}} \quad (*)$

$$\begin{aligned} & g^x \frac{1}{g+h} - \frac{1}{g^{(g+h)}} \\ & \frac{g - (g+h)}{g(g+h)} \end{aligned} \quad \text{COMPLETE:} \quad \begin{aligned} & b^x \frac{1}{a} + \frac{b^{x+a}}{b^x a} = \frac{b}{ba} + \frac{a}{b \cdot a} \\ & = \frac{b+a}{ba} \end{aligned}$$

$$= \frac{-h}{g(g+h)} \quad \text{L.O.G.O.,}$$

$$\lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \frac{-h}{g(g+h)}$$

$$= \lim_{h \rightarrow 0} \left(\frac{-\cancel{h}}{g(g+h)} \cdot \frac{1}{\cancel{h}} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{-1}{g(g+h)} \right) = -\frac{1}{81}$$

33 $\lim_{x \rightarrow 9} \frac{x^2 - 81}{3 - \sqrt{x}}$

$$f(x) = \frac{x^2 - 9^2}{3 - \sqrt{x}}$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$= \frac{(x+9)(x-9)}{3-\sqrt{x}} = \frac{(x+9)(\sqrt{x}+3)(\cancel{\sqrt{x}-3})}{-(\cancel{\sqrt{x}-3})} = \frac{(x+9)(\sqrt{x}+3)}{-1}$$

Logo,

$$\lim_{x \rightarrow 9} \left(\frac{(x+9)(\sqrt{x}+3)}{-1} \right) = \frac{(9+9) \cdot (\sqrt{9}+3)}{-1}$$
$$= \frac{18(3+3)}{-1} = -108 //$$